

SINGULAR OPTIMAL STOCHASTIC CONTROLS I: EXISTENCE*

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Abstract. We apply the compactification method to study the control problem where the state is governed by an Ito stochastic differential equation allowing both classical and singular control. The problem is reformulated as a martingale problem on an appropriate canonical space after the relaxed form of the classical control is introduced. Under some mild continuity hypotheses on the data, it is shown by purely probabilistic arguments that an optimal control for the problem exists. The value function is shown to be Borel measurable.

Key words. singular controls, relaxed controls, control rules, existence theory, pseudopath topology, compactification method, value function

AMS subject classifications. 49J30, 49A55, 60G44, 93E20

1. Introduction. The class of singular stochastic control problems, which has been studied extensively in recent years, deals with systems described by a stochastic differential equation in which one restricts the cumulative displacement of the state caused by control to be of an additive nature. More precisely, in this paper we study the existence of optimal controls to the problem in which the state evolves according to the d -dimensional stochastic differential equation

$$x_t = x + \int_s^t b(\theta, x_\theta, u_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, u_\theta) dB_\theta + \int_s^t g(\theta) dv_\theta$$

on some filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, where $b(\cdot, \cdot, \cdot)$, $\sigma(\cdot, \cdot, \cdot)$, $g(\cdot)$ are given deterministic functions, $(B_t, t \geq 0)$ is a d -dimensional Brownian motion (in fact B need not be d -dimensional), x is the initial state at time s , and $u : [0, T] \mapsto U$, $v : [0, T] \mapsto \mathbb{R}^k$ with v nondecreasing componentwise, stand for the controls.

The expected cost has the form

$$J(\alpha) \equiv E^P \left\{ \int_s^T f(t, x_t, u_t) dt + \int_{[s, T)} c(t) \cdot dv_t \right\},$$

where $f(\cdot, \cdot, \cdot) : [0, T] \times \mathbb{R}^d \times U \mapsto \mathbb{R}$, $c(\cdot) : [0, T] \mapsto \mathbb{R}_+^k$ are given. We assume that the cost of applying the singular control is positive, i.e., $c^i(\cdot) > 0$, $i = 1, \dots, k$.

Some special cases of the one-dimensional problem of this type (without the *classical control* u) have been studied by many authors including Beněš, Shepp, and Witsenhausen [2]; Chow, Menaldi, and Robin [6]; El Karoui and Karatzas [8]; Harrison and Taksar [11]; Karatzas [19], [20]; Karatzas and Shreve [21]; Lehoczky and Shreve [23]; Ma [24]; Menaldi and Robin [25], [26]; and Sun [32]. It is shown that the value function satisfies a variational inequality which gives rise to a free-boundary problem, and the optimal state process is a diffusion reflected at the free boundary. This approach encounters substantial difficulties for the problems in high dimension

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due to the lack of information about the regularity of the associated free boundary. In Soner and Shreve [30], a special two-dimensional problem ($b = 0$, $\sigma = I$) was considered. It was shown there that the associated free boundary is smooth enough to construct a reflected diffusion in the continuation region. However, as the authors pointed out, the method depends heavily on the special features of the problem and cannot be extended to general problems. Another result about high-dimensional problems can be found in Menaldi and Taksar [27], who considered the n -dimensional case with $b = \text{const}$, $\sigma = \text{const}$. It was shown that the value function satisfies the associated Hamilton–Jacobi–Bellman equation, and the existence of the optimal control was proved without requiring any regularity about the free boundary.

In this paper, we apply the compactification method used in Haussmann [12], Haussmann and Lepeltier [13], and El Karoui, Nguyen, and Jeanblanc-Picqué [7] to show the existence of the optimal control. An advantage of this approach is that it does not require any regularity of the value function and thus needs only very mild hypothesis on the data. Unfortunately, it does not show any particular structure of the optimal control.

This paper is organized as follows: In §2 the concept of relaxed control (for the classical control u) is introduced and the problem is reformulated as a martingale problem on a canonical space. In §2.3 a topology is given to the canonical space to make it a metrizable separable space, which is necessary to apply Prohorov’s theorem. A relative compactness criterion for the subsets of the canonical space is given. In §3 it is proved that the optimal control exists, and the value function is shown to be Borel measurable. We also make some comments about possible generalizations of the model and constraints on the state in §4.

This paper is one of a series that applies the direct method to study the singular control problems. In [14] we prove the abstract form of the dynamic programming principle under the same conditions on the data assumed in this paper. When assuming Lipschitz continuity of the data, it is shown that the value function is continuous and is the unique viscosity solution of the corresponding Hamilton–Jacobi–Bellman equation. This method can also be used to show (cf. [33]) that there is a region A such that the optimal control yields no jump when the state is in A , and there exists an optimal control to keep the state in the closure of A after a possible initial jump. In [15] the adaptedness of the optimal control to the state is considered.

Note that in the literature *singular control* usually means that there is no classical control u involved. But our model obviously includes this case by letting U be a singleton (or by letting σ, b, f be independent of u). It also includes the monotone follower as well as the bounded variation problem (cf. §4).

Finally, we list some notation that will be used throughout this paper:

- $\mathbb{R}^d, \mathbb{R}$ denote the d -dimensional Euclidean space and the real line, respectively. $\mathbb{R}_+ = \{x \in \mathbb{R}, x \geq 0\}$, and $\mathbb{R}_+^d$ is defined similarly. For $x = (x^i), y = (y^i) \in \mathbb{R}^d$, $x \cdot y = \sum_{i=1}^d x^i y^i$.
- $T > 0$ is the fixed horizon, and $\Sigma = [0, T] \times \mathbb{R}^d$.
- $C[0, T]$ denotes the collection of real-valued continuous functions defined on $[0, T]$, and $C^d[0, T]$ denotes the collection of $\mathbb{R}^d$ -valued continuous functions defined on $[0, T]$.
- $\mathcal{D}^d[0, T]$ denotes the collection of $\mathbb{R}^d$ -valued functions defined on $[0, T]$ that are left continuous and have right limits (i.e., *lcrl* functions).
- $\mathcal{A}^k[0, T]$ denotes the collection of functions $a : [0, T] \mapsto \mathbb{R}^k$ such that $a = (a^i) \in \mathcal{D}^k[0, T]$ and a^i is nondecreasing with $a^i(0) = 0$, $i = 1, \dots, k$.

- $\mathcal{S}^{l \times k}$ is the space of $l \times k$ matrices with the $l \times k$ -dimensional Euclidean norm.
- If Y is a metric space, $\mathcal{B}(Y)$ denotes the corresponding Borel σ -field, and $f \in \mathcal{B}(Y)$ means that f is a $\mathcal{B}(Y)$ -measurable real-valued function. We denote by $\mathcal{M}_+(Y)$, $\mathcal{M}_1(Y)$ the space of nonnegative Radon measures and the space of probabilities on Y , respectively.
- If X is a random variable on a probability space $(\Omega, \mathcal{F}, P)$, the expectation of X will be denoted by $E^P(X)$. $\mathcal{M}_2^c(\mathcal{M}_2^{c, \text{loc}})$ is the family of continuous square integrable martingales (local martingales, respectively) on some given probability space $(\Omega, \mathcal{F}, P)$ with a filtration $\{\mathcal{F}_t\}$.

2. Formulation of the problem. We consider the following optimal control problem in which we allow both classical control and singular control to act at the same time. The dynamics are in the form

$$(2.1) \quad x_t = x + \int_s^t b(\theta, x_\theta, u_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, u_\theta) dB_\theta + \int_s^t g(\theta) dv_\theta \quad \text{a.s.}$$

for $(s, x) \in \Sigma$, $s \leq t \leq T$, where

- $u_\theta \in U$, $s \leq \theta \leq T$, and U , called the control set, is a compact metric space;
- $(\sigma, b) : \Sigma \times U \mapsto \mathcal{S}^{d \times d} \times \mathbb{R}^d$, $g : [0, T] \mapsto \mathcal{S}^{d \times k}$; $\sigma(t, x, u)$, $b(t, x, u)$ are bounded, measurable, and continuous with respect to (x, u) ; $g(t)$ is continuous on $[0, T]$;
- $(B_t, 0 \leq t \leq T)$ is a d -dimensional Brownian motion on some probability space;
- $v \in \mathcal{A}^k[0, T]$.

We introduce the concept of controls to the stochastic differential equation (2.1).

DEFINITION 2.1. A control is a term $\alpha = (\Omega, \mathcal{F}, \mathcal{F}_t, P, B_t, x_t, u_t, v_t, s, x)$ such that

- (1) $(s, x) \in \Sigma$;
- (2) $(\Omega, \mathcal{F}, P)$ is a probability space with the filtration $\{\mathcal{F}_t\}_{t \geq 0}$;
- (3) u_t is a U -valued process, progressively measurable with respect to $\{\mathcal{F}_t\}_{t \geq 0}$;
- (4) v is $\mathbb{R}_+^k$ -valued processes, progressively measurable with respect to $\mathcal{F}_t$; the sample paths of v are in $\mathcal{A}^k[0, T]$, i.e., for each $\omega \in \Omega$, $v(\omega) \in \mathcal{A}^k[0, T]$;
- (5) B_t is a standard d -dimensional Brownian motion on $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$ and x_t , the state process, is $\mathcal{F}_t$ -adapted with sample paths in $\mathcal{D}^d[0, T]$, and such that (2.1) is satisfied. We assume that $x_r = x$ for $0 \leq r \leq s$.

We call (s, x) the initial condition of the control α .

The collection of controls with initial condition (s, x) is denoted by $\Lambda_{s, x}$. It is well known from the theory of stochastic differential equations that, under the above conditions, the set $\Lambda_{s, x}$ is nonempty for each fixed (s, x) (e.g., take u and v to be constants). The cost corresponding to the control α is defined in the form

$$(2.2) \quad J(\alpha) \equiv E^P \left\{ \int_s^T f(t, x_t, u_t) dt + \int_{[s, T)} c(t) \cdot dv_t \right\},$$

where

- $f : \Sigma \times U \mapsto \mathbb{R}$ is a measurable function and is lower semicontinuous in (x, u) , satisfying

$$-K \leq f(t, x, u) \leq C(1 + \|x\|^m), \quad (t, x, u) \in \Sigma \times U$$

for some constants $m \geq 0$, $K \geq 0$, and $C \geq 0$;

• $c = (c^i) : [0, T] \mapsto \mathbb{R}^k$ is lower semicontinuous and $c^i > 0$, $1 \leq i \leq k$.
Throughout this paper, we write

$$\int_s^t k(\theta) \cdot da(\theta) = \sum_{i=1}^d \int_{[s,t)} k_i(\theta) da_i(\theta)$$

for any $\mathbb{R}^k$ -valued Borel measurable functions $k = (k_i)$ and $a = (a_i) \in \mathcal{A}^k[0, T]$. For $v \in \mathcal{A}^k[0, T]$, define

$$(2.3) \qquad G_t(v) = \int_{[s,t)} g(\theta) dv(\theta), \quad s < t \leq T,$$

and $G_t(v) = 0$ if $0 \leq t \leq s$. It can be verified easily that $G \cdot (v) \in \mathcal{D}^d[0, T]$.
The value function of the problem is defined, for $(s, x) \in \Sigma$, as

$$(2.4) \qquad W(s, x) = \inf_{\alpha \in \Lambda_{s,x}} J(\alpha).$$

A control $\alpha^* \in \Lambda_{s,x}$ is called an *optimal control* if $W(s, x) = J(\alpha^*)$.

Remark 2.2. Note that there is no terminal cost in our model. Because of the way we choose the topology on the canonical space, this method cannot treat the case with terminal cost. We will return to this point at the end of the paper.

2.1. Relaxed controls. In order to apply the compactification method, we now reformulate the problem. Since the Brownian motion in the definition of controls is unknown in advance, we can reformulate the above control problem as an equivalent martingale problem. This simplifies taking limits. In fact, let

$$\mathcal{L} \equiv \frac{1}{2} \sum_{i,j} a_{ij} \frac{\partial^2}{\partial x_i \partial x_j} + \sum_i b_i \frac{\partial}{\partial x_i},$$

where $a(t, x, u) = \sigma(t, x, u)\sigma(t, x, u)^*$. Then we can show that $\alpha \in \Lambda_{s,x}$ if and only if α satisfies (1)–(4) in Definition 2.1, and

- (5') x_t is an $\mathcal{F}_t$ -adapted process with sample paths in $\mathcal{D}^d[0, T]$ such that
- $P(x_r = x, u_r = u^0, v_r = 0, 0 \leq r \leq s) = 1$, where $u^0 \in U$ is arbitrary but fixed,
 - $\forall \phi \in C_b^2(\mathbb{R}^d)$, $\mathcal{M}\phi \in \mathcal{M}_b^c$, i.e., $M_t\phi$ ($0 \leq t \leq T$) is a continuous square integrable martingale on the filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, where

$$\begin{aligned} M_t\phi(\omega) \equiv & \phi(x_t) - \int_s^t \mathcal{L}\phi(\theta, x_\theta, u_\theta) d\theta - \int_s^t \nabla_x \phi(x_\theta) \cdot g(\theta) dv_\theta \\ & - \sum_{s \leq \theta < t} [\phi(x_{\theta+}) - \phi(x_\theta) - \nabla_x \phi(x_\theta) \cdot \Delta x_\theta]. \end{aligned}$$

Therefore, we can delete the term B_t from the notation of a control. The proof of the equivalence of the existence of weak solutions to a stochastic differential equation and the existence of solutions to the corresponding martingale problem, given by Ikeda and Watanabe in Proposition IV-2.1 of [16], also works here despite the extra term $G \cdot (v)$.

Next we introduce the concept of relaxed controls, which gives a more suitable topological structure when applying the compactification method. In a relaxed control

problem, the U -valued process $\{u_t\}$ is replaced by an $M_1(U)$ -valued process $\{\mu_t\}$, where $M_1(U)$ is the space of probability measures on U endowed with the topology of weak convergence. $M_1(U)$ is also a compact metrizable space. If $\phi : U \mapsto \mathbb{R}$ is a bounded measurable function, then we extend ϕ to $M_1(U)$ by letting

$$\phi(\mu) \equiv \int_U \phi(u) \mu(du).$$

DEFINITION 2.3. $\alpha = (\Omega, \mathcal{F}, \mathcal{F}_t, P, x_t, \mu_t, v_t, s, x)$ is called a relaxed control if it satisfies conditions (1), (2), and (4) in Definition 2.1, and

- (3') μ_t is $M_1(U)$ -valued, progressively measurable with respect to $\{\mathcal{F}_t\}_{t \geq 0}$;
 (5'') x_t is an $\mathcal{F}_t$ -adapted process with sample paths in $\mathcal{D}^d[0, T]$ such that
- $P(x_r = x, \mu_r = \delta^0, v_r = 0, 0 \leq r \leq s) = 1$, where δ^0 is the Dirac measure at some arbitrary but fixed $u^0 \in U$;
 - $\forall \phi \in C_b^2(\mathbb{R}^d)$, $\mathcal{M}\phi \in \mathcal{M}_2^c$, i.e., $M_t\phi$ ($0 \leq t \leq T$) is a continuous square integrable martingale on the filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, where

$$\begin{aligned} M_t\phi(\omega) \equiv & \phi(x_t) - \int_s^t \mathcal{L}\phi(\theta, x_\theta, \mu_\theta) d\theta - \int_s^t \nabla_x \phi(x_\theta) \cdot g(\theta) dv_\theta \\ & - \sum_{s \leq \theta < t} [\phi(x_{\theta+}) - \phi(x_\theta) - \nabla_x \phi(x_\theta) \cdot \Delta x_\theta]. \end{aligned}$$

Note that we never work with $\sigma(t, x, u)$. Instead, we will be working on $a(t, x, u)$ through the martingale formulation in Definition 2.3. It is not true in general that $\sigma(t, x, \mu)\sigma(t, x, \mu)^* = a(t, x, \mu)$ (cf. El Karoui, Nguyen, and Jeanblanc-Piqué [7]).

The collection of relaxed controls starting from time s with the initial state x is denoted by $\tilde{\Lambda}_{s,x}$. Note that $\Lambda_{s,x}$ can be imbedded into $\tilde{\Lambda}_{s,x}$ by letting $\mu_t(du) \equiv \delta_{u_t}(du)$. Here, δ_u denotes the Dirac measure at the point u . Hence, $\forall \alpha \in \Lambda_{s,x}$, there exists an $\tilde{\alpha} \in \tilde{\Lambda}_{s,x}$ such that

$$J(\tilde{\alpha}) = J(\alpha),$$

and, therefore,

$$\inf_{\tilde{\alpha} \in \tilde{\Lambda}_{s,x}} J(\tilde{\alpha}) \leq \inf_{\alpha \in \Lambda_{s,x}} J(\alpha).$$

In order to get the reverse inequality, we define for each $(t, x) \in \Sigma$ a subset of $S^{d \times d} \times \mathbb{R}^d \times \mathbb{R}$:

$$(2.5) \quad K(t, x) \equiv \{(a(t, x, u), b(t, x, u), z) : z \geq f(t, x, u), u \in U\}.$$

PROPOSITION 2.4. Assume that $K(t, x)$ is convex $\forall (t, x) \in \Sigma$. Then $\forall \tilde{\alpha} \in \tilde{\Lambda}_{s,x}$, there exists an $\alpha \in \Lambda_{s,x}$ such that

$$J(\alpha) \leq J(\tilde{\alpha}).$$

The proof in Haussmann and Lepeltier [13, Thm. 3.6.], can be carried over here without any change. From this result we know that when $K(t, x)$ is convex,

$$\inf_{\tilde{\alpha} \in \tilde{\Lambda}_{s,x}} J(\tilde{\alpha}) = \inf_{\alpha \in \Lambda_{s,x}} J(\alpha).$$

Hence we can restrict our attention to the case of relaxed controls. Moreover, if the infimum over relaxed controls is attained, then so is the infimum over ordinary controls.

If the convexity condition fails, our existence results pertain only to relaxed controls.

2.2. Control rules. Now we choose a canonical space to simplify the arguments in the compactification method. Define

$$\mathcal{U} \equiv \{\mu : [0, T] \mapsto M_1(U) \text{ is Borel measurable}\}.$$

Consider the canonical space

$$\mathcal{X} \equiv \mathcal{D}^d[0, T] \times \mathcal{U} \times \mathcal{A}^k[0, T].$$

All the above spaces are equipped with appropriate topologies, which we will discuss later. Let $\tilde{\mathcal{D}}, \tilde{\mathcal{U}}, \tilde{\mathcal{A}}$ denote their Borel σ -fields, and $\tilde{\mathcal{D}}_t, \tilde{\mathcal{U}}_t, \tilde{\mathcal{A}}_t$ denote the σ -fields up to time t , e.g.,

$$\tilde{\mathcal{A}}_t = \sigma\{v(\theta) : 0 \leq \theta \leq t\}, \quad 0 \leq t \leq T.$$

Let

$$\begin{aligned} \tilde{\mathcal{X}} &\equiv \tilde{\mathcal{D}} \times \tilde{\mathcal{U}} \times \tilde{\mathcal{A}}, \\ \tilde{\mathcal{X}}_t &\equiv \tilde{\mathcal{D}}_t \times \tilde{\mathcal{U}}_t \times \tilde{\mathcal{A}}_t. \end{aligned}$$

DEFINITION 2.5. A control rule is a probability R on the measurable space $(\mathcal{X}, \tilde{\mathcal{X}})$ such that

$$\tilde{\alpha} = (\mathcal{X}, \tilde{\mathcal{X}}, \tilde{\mathcal{X}}_t, R, x_t, \mu_t, v_t, s, x)$$

is a relaxed control, where

$$x_t(\omega) = x_t, \quad \mu_t(\omega) = \mu_t, \quad v_t = v_t(\omega)$$

for $\omega = (x, \mu, v) \in \mathcal{X}$, i.e.,

(1) $(\mathcal{X}, \tilde{\mathcal{X}}, R)$ is a probability space with the filtration $\{\tilde{\mathcal{X}}_t\}$, $(s, x) \in \Sigma$, and

$$R(x_r = x, \mu_r = \delta^0, v_r = 0, 0 \leq r \leq s) = 1;$$

(2) $\forall \phi \in C_b^2(\mathbb{R}^d)$, $M_t \phi$ ($s \leq t \leq T$) is in $\mathcal{M}_2^c$ on the filtered probability space $(\mathcal{X}, \tilde{\mathcal{X}}, \tilde{\mathcal{X}}_t, R)$ ($s \leq t \leq T$), where

$$\begin{aligned} M_t \phi(\omega) &\equiv \phi(x_t) - \int_s^t \mathcal{L} \phi(\theta, x_\theta, \mu_\theta) d\theta - \int_s^t \nabla_x \phi(x_\theta) \cdot g(\theta) dv_\theta \\ &\quad - \sum_{s \leq \theta < t} [\phi(x_{\theta+}) - \phi(x_\theta) - \nabla_x \phi(x_\theta) \cdot \Delta x_\theta]. \end{aligned}$$

Let $J(s, R) \equiv J(\tilde{\alpha})$.

We denote by $\mathcal{R}_{s,x}$ the space of control rules such that the above relaxed control starts from time s with initial state x . We will suppress s in $J(s, R)$ when it is clear that $R \in \mathcal{R}_{s,x}$. Now the control problem can be described completely in terms of control rules.

PROPOSITION 2.6. Let $\alpha = (\Omega, \mathcal{F}, \mathcal{F}_t, P, x_t, \mu_t, v_t, s, x)$ be a relaxed control; then there exists a control rule $R \in \mathcal{R}_{s,x}$ such that

$$J(R) = J(\alpha).$$

Proof. The proof of this result is standard. Define a map $\Phi : \Omega \mapsto \mathcal{X}$ by

$$\Phi(\omega) \equiv (x.(\omega), \mu.(\omega), v.(\omega)).$$

This map is measurable, and $\mathcal{F}_t^{x, \mu, v} \subset \Phi^{-1}(\tilde{\mathcal{X}}_t) \subset \mathcal{F}_t$. We can show that

$$\tilde{\alpha} = (\Omega, \mathcal{F}, \Phi^{-1}(\tilde{\mathcal{X}}_t), P, x_t, \mu_t, v_t, s, x)$$

is also a relaxed control such that

$$J(\tilde{\alpha}) = J(\alpha).$$

Let $R = P \circ \Phi^{-1}$, which is a probability on $(\mathcal{X}, \tilde{\mathcal{X}})$. It is easily seen that

$$(\mathcal{X}, \tilde{\mathcal{X}}, \tilde{\mathcal{X}}_t, R, x_t, \mu_t, v_t, s, x)$$

is a relaxed control satisfying the requirements in the Proposition. $\square$

2.3. The topology on the canonical space. In this section we define the topologies on the spaces $\mathcal{D}^d[0, T]$, $\mathcal{U}$, and $\mathcal{A}^k[0, T]$.

The space $\mathcal{D}^d[0, T]$. We first give a topology on $\mathcal{D}^d[0, \infty)$, the collection of *lcrl* functions on $\mathbb{R}_+$ taking values in $\mathbb{R}^d$. Then we can take $\mathcal{D}^d[0, T]$ as a subset of $\mathcal{D}^d[0, \infty)$ by extending each $x \in \mathcal{D}^d[0, T]$ to an $x' \in \mathcal{D}^d[0, \infty)$ through

$$x'(t) = \begin{cases} x(t), & 0 \leq t < T, \\ x(T), & t \geq T, \end{cases}$$

and consider the induced topology on $\mathcal{D}^d[0, T]$.

Define a measure $\lambda(\cdot)$ on the Borel subsets of $\mathbb{R}_+$ by $\lambda(dt) = e^{-t}dt$. For a Borel-measurable function $f : \mathbb{R}_+ \mapsto \mathbb{R}^d$, the image of the measure $\lambda(\cdot)$ under the mapping $t \mapsto (t, f(t))$ is called the *pseudopath* of the function f . It is a probability law on $[0, \infty) \times \bar{\mathbb{R}}^d$ and is denoted by $\Psi(f)$. It is clear that Ψ identifies two functions if and only if they are equal almost everywhere in the Lebesgue sense and, in particular, Ψ is one-to-one on $\mathcal{D}^d[0, \infty)$. Thus it provides us with an imbedding of $\mathcal{D}^d[0, \infty)$ into the compact Polish space $\bar{\mathcal{P}}$ of all the probabilities on $[0, \infty) \times \bar{\mathbb{R}}^d$ (with the topology of weak convergence). The topology that $\bar{\mathcal{P}}$ induces on $\mathcal{D}^d[0, \infty)$ via the mapping Ψ is called the *pseudopath topology*, and makes $\mathcal{D}^d[0, \infty)$ a separable metrizable space. The associated Borel σ -algebra on $\mathcal{D}^d[0, \infty)$ is the same one that we get from the Skorohod topology. In fact, Lemma 1 in [28] tells us that convergence in the pseudopath topology is just convergence in measure. Let $\tilde{\mathcal{D}}$ denote the Borel σ -field on $\mathcal{D}^d[0, T]$, and $\tilde{\mathcal{D}}_t$ be the Borel σ -field up to time t , i.e.,

$$\tilde{\mathcal{D}}_t = \sigma\{x(\theta) : 0 \leq \theta \leq t\}, \quad 0 \leq t \leq T.$$

Now we state a relative compactness criterion for subsets of $\mathcal{D}^d[0, \infty)$. For $x \in \mathcal{D}^d[0, \infty)$, define $x^* = \sup_t \|x(t)\|$. For $u = (u^i), v = (v^i) \in \mathbb{R}^d$, $u < v$ means $u^i < v^i$ ($1 \leq i \leq d$). Let $N^{uv}(x) = \sum_1^d N^{u^i v^i}(x^i)$, where $N^{u^i v^i}(x^i)$ denotes the number of upcrossings of $x^i(\cdot)$ on $[0, \infty)$ between the levels u^i and v^i . Then a subset $A \subset \mathcal{D}^d[0, \infty)$ such that

$$(2.6) \quad \sup_{x \in A} x^* < \infty, \quad \sup_{x \in A} N^{uv}(x) < \infty$$

for any $u < v$ is relatively compact in $\mathcal{D}^d[0, \infty)$ with the pseudopath topology. For details, see Meyer and Zheng [28].

Remark 2.7. As pointed out by Meyer and Zheng [28], $\mathcal{D}^d[0, \infty)$ with the pseudopath topology is not a Polish space. But from the definition we know that it is homeomorphic to a subspace of the Polish space $\bar{\mathcal{P}}$, and hence is a separable metric space. $\square$

The space $\mathcal{U}$. $\mathcal{U}$ is the space of measurable transformations $\mu : [0, T] \mapsto \mathcal{M}_1(U)$ endowed with the stable topology, which is defined as follows: for $A \in \mathcal{B}([0, T])$, $B \in \mathcal{B}(U)$, define

$$\bar{\mu}(A \times B) \equiv \int_A \mu_t(B) dt.$$

Then $\bar{\mu}$ can be extended uniquely to an element in $\mathcal{M}_+([0, T] \times U)$, the space of nonnegative Radon measures on $[0, T] \times U$. The stable topology on $\mathcal{U}$ is the weakest topology that renders continuous the mappings

$$\bar{\mu} \mapsto \int_0^T \int_U \phi(t, u) \bar{\mu}(dt, du)$$

for all bounded measurable functions ϕ that are continuous in u .

Under this topology, we know that $\mathcal{M}_+([0, T] \times U)$ is a compact separable metrizable space. $\mathcal{U}$ is also endowed with its Borel σ -field $\tilde{\mathcal{U}}$, which is the smallest σ -field such that the mappings

$$\mu \mapsto \int_0^T \int_U \mu_t(du) f(t, u) dt$$

are measurable, where f is a bounded measurable function continuous with respect to the variable u . The filtration $\tilde{\mathcal{U}}_t$ is the σ -field generated by $\{1_{[0, t]}\mu, \mu \in \mathcal{U}\}$. From the definition of the stable topology, we know that $\tilde{\mathcal{U}}_t$ is generated by the sets of the form

$$\left\{ \mu : \int_0^s \mu_\theta d\theta \in B \right\}$$

with $s \leq t$ and B a Borel set in $\mathcal{M}_+(U)$.

For more details, see Haussmann and Lepeltier [13].

The space $\mathcal{A}^k[0, T]$. Let $\mathcal{V}$ be the collection of functions $a : [0, T] \mapsto \mathbb{R}$ such that each $a(\cdot)$ is of bounded variation and left continuous. We assume $a(0) = 0$. We first consider a topology on $\mathcal{V}$.

Let $\mathcal{M}[0, T]$ be the collection of signed Radon measures on $[0, T]$. Then there is a one-to-one correspondence between $\mathcal{V}$ and $\mathcal{M}[0, T]$: for $a \in \mathcal{V}$, let

$$\nu_a([s, t)) \equiv a(t) - a(s) \quad \forall 0 \leq s \leq t \leq T.$$

So we need only consider a topology on $\mathcal{M}[0, T]$, then we can get the induced topology on $\mathcal{V}$.

Denote by $C[0, T]$ the collection of real-valued continuous functions on $[0, T]$. It is well known that $\mathcal{M}[0, T]$, and, therefore, $\mathcal{V}$ is the dual space of $C[0, T]$ with the supremum norm: for $f \in C[0, T]$,

$$\|f\| = \sup_{0 \leq t \leq T} |f(t)|,$$

and the corresponding weak* topology on $\mathcal{M}[0, T]$ is the topology induced by the weak convergence of measures. It can be easily seen that the measures of the form $\sum_1^N a_i \delta_{x_i}$ with N finite and a_i, x_i rational comprise a countable dense subset of $\mathcal{M}[0, T]$; therefore, $\mathcal{M}[0, T]$, with the weak* topology, is separable.

Let $\mathcal{A}^0 = \{a \in \mathcal{V} : a \text{ nondecreasing}\}$. Then it is a closed subset of $\mathcal{V}$ under the weak convergence topology, and the corresponding closed subset in $\mathcal{M}[0, T]$ is $\mathcal{M}_+[0, T]$. By Stroock and Varadhan [31] we know that $\mathcal{M}_+[0, T]$ is a metrizable space. Therefore, we can conclude that under the weak convergence topology $\mathcal{A}^0$ is a separable metric space.

We state the following theorem which will be used later.

THEOREM 2.8. *For any constant $C > 0$,*

$$(2.7) \quad \{\lambda \in \mathcal{M}_+[0, T] : \lambda([0, T]) \leq C\}$$

is a compact subset of $\mathcal{M}_+[0, T]$.

Proof. Note that the set (2.7) is a closed subset in $\mathcal{M}[0, T]$ in both the (variation) norm topology and the weak* topology. Also notice that if $\lambda \in \mathcal{M}_+[0, T] \subset \mathcal{M}[0, T]$, then the norm of λ will be

$$\|\lambda\| = \sup_{\|f\| \leq 1} \left| \int_0^T f d\lambda \right| = \lambda([0, T]),$$

i.e., the set (2.7) is a norm-bounded subset of $\mathcal{M}[0, T]$. Therefore by the Banach–Alaoglu theorem (see Larsen [22, Thm. 9.4.1]) we can conclude that (2.7) is a compact subset of $\mathcal{M}[0, T]$, and, therefore, a compact subset of $\mathcal{M}_+[0, T]$. $\square$

Finally, observe that $\mathcal{A}^k[0, T] = (\mathcal{A}^0)^k$ and consider the product topology on $\mathcal{A}^k[0, T]$ inherited from the weak topology of $\mathcal{A}^0$. We can state the following result.

COROLLARY 2.9. *$\mathcal{A}^k[0, T]$ is metrizable and separable. For $v_n, v \in \mathcal{A}^k[0, T]$, $v_n \rightarrow v$ if and only if*

$$\int_0^T f(t) \cdot dv_n(t) \rightarrow \int_0^T f(t) \cdot dv(t)$$

for any $f \in C([0, T], \mathbb{R}^k)$. Moreover, the set

$$(2.8) \quad V_M = \{v \in \mathcal{A}^k[0, T] : \|v(T)\| \leq M\}$$

is compact for any constant $M > 0$.

We write down the following observation for later use. Its proof is obvious from the relative compactness criterion for pseudopath topology on $\mathcal{D}^d[0, T]$. Recall that the map $G : \mathcal{A}^k[0, T] \mapsto \mathcal{D}^d[0, T]$ is defined by (2.3).

LEMMA 2.10. *For any constant $M > 0$*

$$G(V_M) = \{G(v) : v \in V_M\}$$

is a relatively compact subset of $\mathcal{D}^d[0, T]$, where V_M is defined by (2.8).

From now on we will always use the notation $\Omega = \mathcal{X}$, $\mathcal{F} = \tilde{\mathcal{X}}$, and $\mathcal{F}_t = \tilde{\mathcal{X}}_t$. It is well known that $\mathcal{M}_1(\Omega)$, endowed with the Prohorov weak convergence topology, is then a separable metrizable space. Denote the collection of all the control rules with initial condition (s, x) by $\mathcal{R}_{s,x}$, which is a subset of $\mathcal{M}_1(\Omega)$. For any real number λ , define

$$\mathcal{R}_{s,x}^\lambda \equiv \{P \in \mathcal{R}_{s,x} : J(s, P) \leq \lambda\}.$$

PROPOSITION 2.11. *There exists a constant $C \geq 0$ such that for*

$$(2.9) \quad \lambda(x) = C(1 + \|x\|^m), \quad x \in \mathbb{R}^d,$$

we have $\mathcal{R}_{s,x}^{\lambda(x)} \neq \emptyset$ for each $(s, x) \in \Sigma$. Recall that m is given in the definition of f .

Proof. Under our assumptions, it is known from the theory of stochastic differential equations (cf. Stroock and Varadhan [31]) that there exists a $P_0 \in \mathcal{R}_{s,x}$ with

$$P_0(v_t = 0, \mu_t = \delta_{u^0}, 0 \leq t \leq T) = 1.$$

Then from the definition of control rule we know that under P_0 ,

$$x_t = x + \int_s^t b(\theta, x_\theta, u^0) d\theta + \int_s^t \sigma(\theta, x_\theta, u^0) dB_\theta;$$

therefore, from the boundedness of b , σ , and the Burkholder–Davis–Gundy inequality we have

$$\begin{aligned} E^{P_0} \|x_t\|^m &\leq C \left\{ \|x\|^m + E^{P_0} \left\| \int_s^t b(\theta, x_\theta, u^0) d\theta \right\|^m \right. \\ &\quad \left. + E^{P_0} \sup_{0 \leq \theta \leq t} \left\| \int_s^\theta \sigma(\theta', x_{\theta'}, u^0) dB_{\theta'} \right\|^m \right\} \\ &\leq C \left\{ 1 + \|x\|^m + E^{P_0} \left(\int_s^t \text{tr}(a(\theta, x_\theta, u^0)) d\theta \right)^{\frac{m}{2}} \right\} \\ &\leq C(1 + \|x\|^m), \end{aligned}$$

where C is a constant independent of x . Now we have by definition

$$\begin{aligned} J(s, P_0) &= E^{P_0} \left\{ \int_s^T f(\theta, x_\theta, u^0) d\theta \right\} \leq C \left(1 + \int_s^T E^{P_0} \|x_\theta\|^m d\theta \right) \\ &\leq C(1 + \|x\|^m), \end{aligned}$$

where C is independent of s, x . The proposition is thus proved by letting $\lambda(x) = C(1 + \|x\|^m)$. $\square$

3. Existence of optimal controls. In order to show the existence of optimal controls, let us reformulate the problem as follows. Consider the stochastic differential equation (2.1), and let

$$y = x - G(v);$$

then $x_t = y_t + G_t(v)$ ($0 \leq t \leq T$), and (2.1) becomes

$$(3.1) \quad y_t = x + \int_s^t b(\theta, x_\theta, u_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, u_\theta) dB_\theta;$$

this is a continuous process. We use this idea, but in the martingale setting, to replace the definition of a control rule.

PROPOSITION 3.1. *$P \in \mathcal{R}_{s,x}$ if and only if there exists an $\mathcal{F}_t$ -adapted process y such that*

- (1) y is continuous with probability one (w.p. 1) and $P(x = y + G(v)) = 1$;
 (2) $P(x_r = x, \mu = \delta^0, v_r = 0, 0 \leq r \leq s) = 1$;
 (3) $\tilde{M}_t \phi \in \mathcal{M}_2^c$ for every $\phi \in C_b^2(\mathbb{R}^d)$, where

$$(3.2) \quad \tilde{M}_t \phi(\omega) = \phi(y_t(\omega)) - \int_s^t \tilde{\mathcal{L}}\phi(\theta, x_\theta, y_\theta, \mu_\theta) d\theta,$$

and

$$\tilde{\mathcal{L}}\phi(\theta, x, y, u) \equiv \frac{1}{2} \sum_{i,j=1}^d a_{ij}(\theta, x, u) \frac{\partial^2 \phi(y)}{\partial y^i \partial y^j} + \sum_{i=1}^d b_i(\theta, x, u) \frac{\partial \phi(y)}{\partial y^i}.$$

Proof. The proof is comparable to that in Suo [33]. $\square$

Some routine calculations can lead to the following result.

LEMMA 3.2. Assume $P \in \mathcal{R}_{s,x}$. For any $\phi, \psi \in C_b^2(\mathbb{R}^d)$ we have

$$(3.3) \quad \langle \tilde{M}\phi, \tilde{M}\psi \rangle_t = \sum_{i,j=1}^d \int_s^t a_{ij}(\theta, x_\theta, \mu_\theta) \frac{\partial \phi(y_\theta)}{\partial y^i} \frac{\partial \psi(y_\theta)}{\partial y^j} d\theta$$

under probability P . In particular, if we define, for $1 \leq i \leq d$,

$$(3.4) \quad \tilde{M}^i(t) = y^i(t) - x^i - \int_s^t b_i(\theta, x_\theta, \mu_\theta) d\theta,$$

i.e., $\tilde{M}_t \phi$ with $\phi(y) = y^i - x^i$, then $\tilde{M} \in \mathcal{M}_2^{c, \text{loc}}$, and

$$\langle \tilde{M}^i, \tilde{M}^j \rangle_t = \int_s^t a_{ij}(\theta, x_\theta, \mu_\theta) d\theta, \quad 1 \leq i, j \leq d.$$

Recall that $C^d[0, T]$ is the space of $\mathbb{R}^d$ -valued continuous functions on $[0, T]$. We give $C^d[0, T]$ the uniform topology, i.e., for $x, y \in C^d[0, T]$, the distance between x and y is defined by

$$\rho(x, y) \equiv \max_{0 \leq t \leq T} \|x(t) - y(t)\|.$$

This makes $C^d[0, T]$ a Polish space (cf. Billingsley [3]).

For a sequence $P^n \in \mathcal{R}_{s_n, x_n}$ with $(s_n, x_n) \in \Sigma$, the probability law of the process y , defined in Proposition 3.1, under P^n is defined by

$$\tilde{P}^n(C) \equiv P^n(\omega : y(\omega) \in C)$$

for $C \in \tilde{\mathcal{C}}$, where $\tilde{\mathcal{C}}$ is the Borel σ -field of $C^d[0, T]$.

PROPOSITION 3.3. If the sequence (s_n, x_n) is bounded in Σ , then $\{\tilde{P}^n\}$ is relatively compact. Moreover, for any $\varepsilon > 0$, there exists a compact subset $K \subset C^d[0, T]$ such that

$$(3.5) \quad \tilde{P}^n(K) \geq 1 - \varepsilon \quad \forall n.$$

Proof. To show that $\{\tilde{P}^n\}$ is relatively compact, we need only verify the following:

- (a) $\lim_{A \rightarrow \infty} \inf_n P^n(\|y(0)\| \leq A) = 1$, and
 (b) for any $\gamma > 0$,

$$(3.6) \quad \lim_{\delta \downarrow 0} \limsup_n P^n \left[\sup_{\substack{0 \leq s < t \leq T \\ t-s < \delta}} \|y(t) - y(s)\| \geq \gamma \right] = 0.$$

Note that (a) is obvious from the fact $P^n(y(0) = x_n) = 1$. By Billingsley [3, Thm. 12.3], (b) is implied by

$$(3.7) \quad E^{P^n} \|y(t_2) - y(t_1)\|^4 \leq C |t_2 - t_1|^2$$

for any $n \geq 1$, $0 \leq t_1, t_2 \leq T$, where C is a constant.

Recall from the definition of y that $P^n(y(t) = x_n, 0 \leq t \leq s_n) = 1$, and for $t \geq s_n$,

$$y(t) = x_n + \int_{s_n}^t b(\theta, x_\theta, \mu_\theta) d\theta + \tilde{M}_n(t)$$

with $\tilde{M}_n \in \mathcal{M}_2^{c, \text{loc}}$ under P_n , and

$$\langle \tilde{M}_n \rangle_t = \int_{s_n}^t \text{tr}(a(\theta, x_\theta, \mu_\theta)) d\theta, \quad t \geq s_n.$$

It can be easily verified that (3.7) follows from the boundedness of the coefficients $\sigma(\cdot, \cdot, \cdot)$ and $b(\cdot, \cdot, \cdot)$ and the Burkholder–Davis–Gundy inequality.

We have therefore shown that the probability laws of y under P^n are relatively compact. The existence of a compact subset K such that (3.5) holds is a consequence of Prohorov's theorem. $\square$

PROPOSITION 3.4. *If A is a bounded subset of $\mathbb{R}^d$, then*

$$(3.8) \quad \lim_{M \rightarrow \infty} \inf_{\substack{P \in \mathcal{R}_{s,x}^\lambda \\ (s,x) \in [0,T] \times A}} P\{\omega : \|v_T\| \leq M\} = 1,$$

where $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^k$.

Proof. If $P \in \mathcal{R}_{s,x}^\lambda$, then $J(P) \leq \lambda$. Since the function f is assumed to be bounded from below, there exists a constant $K > 0$ such that $f \geq -K$. From the assumption that $c(t)$ is strictly positive and lower semicontinuous, there exists a constant $c_0 > 0$ such that $c^i(t) \geq c_0$ ($1 \leq i \leq k$, $0 \leq t \leq T$). Thus

$$\begin{aligned} \lambda &\geq J(P) = E^P \left\{ \int_0^T f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^T c(\theta) \cdot dv_\theta \right\} \\ &\geq E^P \{-KT + c_0 \|v_T\|\}. \end{aligned}$$

Therefore, $E^P \|v_T\| \leq \bar{K} \equiv (\lambda + KT)/c_0$ and

$$P(\|v_T\| \geq M) \leq \frac{\bar{K}}{M}.$$

The proposition is now obvious. $\square$

Define a function on Ω by

$$(3.9) \quad \Gamma_s(\omega) \equiv \int_s^T f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^T c(\theta) \cdot dv_\theta$$

for $\omega = (x, \mu, v)$. Then we have

LEMMA 3.5. $\Gamma_s(\cdot)$ is lower semicontinuous on Ω , i.e.,

$$(3.10) \quad \liminf_n \Gamma_s(\omega_n) \geq \Gamma_s(\omega)$$

if $\omega_n \rightarrow \omega$ in Ω .

Proof. We show the case when $s = 0$; for general s the proof is similar. We can also assume that $f(t, \cdot, \cdot)$, $c(\cdot)$ are continuous. In fact, since they are lower semicontinuous, we can find sequences of continuous functions $\{f_m(t, \cdot, \cdot)\}$, $\{c_m(\cdot)\}$ such that

$$f_m(t, \cdot, \cdot) \uparrow f(t, \cdot, \cdot), \quad c_m^i(\cdot) \uparrow c^i(\cdot), \quad 1 \leq i \leq k,$$

and thus if the lemma is true for continuous functions, then $\forall m \geq 1$,

$$\begin{aligned} \liminf_n \Gamma_0(\omega_n) &\geq \liminf_n \left\{ \int_0^T f_m(\theta, x_n(\theta), \mu_n(\theta)) d\theta + \int_0^T c_m(\theta) \cdot dv_n(\theta) \right\} \\ &\geq \int_0^T f_m(\theta, x(\theta), \mu(\theta)) d\theta + \int_0^T c_m(\theta) \cdot dv(\theta). \end{aligned}$$

Let $m \rightarrow \infty$, and use the monotone convergence theorem to conclude the result.

We assume that $f(t, \cdot, \cdot)$ and $c(\cdot)$ are continuous. From $v_n \rightarrow v$ in $\mathcal{A}^k[0, T]$ we have

$$\int_0^T c(\theta) \cdot dv_n(\theta) \rightarrow \int_0^T c(\theta) \cdot dv(\theta).$$

Now we show that as $n \rightarrow \infty$,

$$(3.11) \quad \int_0^T \int_U f(\theta, x_n(\theta), u) \mu_\theta^n(du) d\theta \rightarrow \int_0^T \int_U f(\theta, x(\theta), u) \mu_\theta(du) d\theta.$$

Let $dQ^n \equiv \mu_\theta^n(du) d\theta$, $dQ \equiv \mu_\theta(du) d\theta$; then the right-hand side of (3.11) can be rewritten as

$$\begin{aligned} (3.12) \quad \int_{[0, T] \times U} f(\theta, x_n(\theta), u) dQ^n &= \int_{[0, T] \times U} f(\theta, x(\theta), u) dQ^n \\ &\quad + \int_{[0, T] \times U} [f(\theta, x_n(\theta), u) - f(\theta, x(\theta), u)] dQ^n. \end{aligned}$$

From the definition of stable topology we know as $n \rightarrow \infty$,

$$\int_{[0, T] \times U} f(\theta, x(\theta), u) dQ^n \rightarrow \int_{[0, T] \times U} f(\theta, x(\theta), u) dQ.$$

Now we show that the limit of the second integral in (3.12) is zero. For any positive integer m and constant $\gamma > 0$, let

$$\begin{aligned} g(t, x, u) &= f(t, x, u) - f(t, x(t), u), \\ A_m &= \left\{ (t, u) : \sup_{\|x - x(t)\| \leq 1/m} |g(t, x, u)| \geq \gamma \right\}. \end{aligned}$$

Then each t -section of the set A_m is a closed subset of U from the continuity of the function $f(t, \cdot, \cdot)$. Also $A_1 \supset A_2 \supset \cdots \supset A_m \supset A_{m+1} \supset \cdots$, and

$$\bigcap_m A_m = \emptyset.$$

Applying the results in Jacod and Mémin [17, Prop. 2.11] we can get

$$(3.13) \quad \limsup_n Q_n(A_m) \leq Q(A_m), \quad \lim_m Q(A_m) = 0.$$

Let $B_n = \{(t, u) : |g(t, x_n(t), u)| \geq \gamma\}$. In order to show that the limit of the last integral in (3.12) is zero, we need only show

$$(3.14) \quad \lim_{n \rightarrow \infty} Q_n(B_n) = 0$$

by Jacod and Mémin [17, Cor. 2.18]. For a given $\varepsilon > 0$, from (3.13) there exists an $M > 0$ such that $Q(A_M) \leq \varepsilon$. Recall that convergence in pseudopath topology is equivalent to the convergence in Lebesgue measure, therefore, $x_n(\cdot) \rightarrow x(\cdot)$ in the Lebesgue measure l , and there exists N such that when $n \geq N$,

$$l \left\{ t : \|x_n(t) - x(t)\| > \frac{1}{M} \right\} < \varepsilon.$$

Let $C_n^M = \{t : \|x_n(t) - x(t)\| > 1/m\}$; then it is obvious that

$$B_n \setminus (C_n^M \times U) \subset A_M,$$

and, therefore, we have

$$Q_n(B_n) \leq Q_n(A_M) + Q_n(C_n^M \times U).$$

But $Q_n(C_n^M \times U) = l(C_n^M) < \varepsilon$, hence

$$\begin{aligned} \limsup_n Q_n(B_n) &\leq \limsup_n Q_n(A_M) + \varepsilon \\ &\leq Q(A_M) + \varepsilon < 2\varepsilon. \end{aligned}$$

Since ε is arbitrary, we have shown (3.14), and the lemma is thus proved. $\square$

For $P \in \mathcal{R}_{s,x}$, we have by definition that

$$J(s, P) = E^P \Gamma_s,$$

i.e., the cost corresponding to the control rule P . We can now state the following result.

THEOREM 3.6. *The mapping $(s, x, P) \rightarrow J(s, P)$ is lower semicontinuous on $\{(s, x, P) : (s, x) \in \Sigma, P \in \mathcal{R}_{s,x}\}$ with the induced topology of $[0, T] \times M_+(\Omega)$, i.e., if $(s_n, x_n) \in \Sigma$, $P_n \in \mathcal{R}_{s_n, x_n}$, $s_n \rightarrow s$, $x_n \rightarrow x$, and $P_n \rightarrow P \in \mathcal{R}_{s,x}$ weakly, then*

$$(3.15) \quad J(s, P) \leq \liminf_n J(s_n, P_n).$$

Proof. Assume $(s_n, x_n, P_n) \rightarrow (s, x, P)$ with $P_n \in \mathcal{R}_{s_n, x_n}$, $P \in \mathcal{R}_{s, x}$. It suffices to consider two cases: $s_n \uparrow s$ or $s_n \downarrow s$. When $s_n \uparrow s$,

$$\begin{aligned} E^{P_n}(\Gamma_{s_n} - \Gamma_s) &= E^{P_n} \left\{ \int_{s_n}^s f(\theta, x_\theta, \mu_\theta) d\theta + \int_{s_n}^s c(\theta) \cdot dv_\theta \right\} \\ &\geq E^{P_n} \left(\int_{s_n}^s f(\theta, x_\theta, \mu_\theta) d\theta \right) \geq -K(s - s_n), \end{aligned}$$

because we have assumed that $c \geq 0$. It follows that

$$(3.16) \quad \liminf_n E^{P_n}(\Gamma_{s_n} - \Gamma_s) \geq 0.$$

When $s_n \downarrow s$, (3.16) follows from

$$E^{P_n}(\Gamma_{s_n} - \Gamma_s) = \int_s^{s_n} f(\theta, x_n, u^0) d\theta,$$

where we have used the fact that $P_n(v_\theta = 0, 0 \leq \theta \leq s_n) = 1$.

In either case, from (3.16) and the lower semicontinuity of $\Gamma_s(\cdot)$, we have

$$\begin{aligned} \liminf_n J(s_n, P_n) &\geq \liminf_n E^{P_n} \Gamma_s + \liminf_n E^{P_n}(\Gamma_{s_n} - \Gamma_s) \\ &\geq \liminf_n E^{P_n} \Gamma_s \geq E^P \Gamma_s = J(s, P), \end{aligned}$$

i.e., $J(\cdot, \cdot)$ is lower semicontinuous. $\square$

THEOREM 3.7. *For any $\lambda > 0$, if A is a compact set in $\mathbb{R}^d$, then $\cup\{\mathcal{R}_{s,x}^\lambda : s \in [0, T], x \in A\}$ is compact.*

Proof. Since $M_1(\Omega)$ is metrizable, we need only to show that each sequence $\{P^n\} \subset \mathcal{R}_{s_n, x_n}^\lambda$ has a subsequence $\{P^{n_k}\}$ such that $P^{n_k} \rightarrow P \in \mathcal{R}_{s,x}^\lambda$ for some $s \in [0, T]$, $x \in A$. Because A is compact and $T < \infty$, we may assume $s_n \rightarrow s$, $x_n \rightarrow x$ for some $s \in [0, T]$, $x \in A$.

By Proposition 3.1, the process y , defined by

$$y.(\omega) = x.(\omega) - G.(v(\omega))$$

for $\omega = (x, \mu, v)$, has continuous sample paths under the probability P^n and is such that $\tilde{M}_t \phi \in \mathcal{M}_2^c$ ($\tilde{M}_t \phi$ is defined in Proposition 3.1) for each $\phi \in C_b^2(\mathbb{R}^d)$. Now we introduce the following auxiliary space:

$$\begin{aligned} \mathcal{Z} &\equiv \Omega \times C^d[0, T], \\ \tilde{\mathcal{Z}} &\equiv \mathcal{F} \times \tilde{\mathcal{C}}. \end{aligned}$$

Define a probability $\tilde{P}^n$ on $(\mathcal{Z}, \tilde{\mathcal{Z}})$ by the probability law of (x, μ, v, y) with respect to P^n , i.e., for $Z \in \tilde{\mathcal{Z}}$,

$$\tilde{P}^n(Z) \equiv P^n(\omega : (x.(\omega), \mu.(\omega), v.(\omega), y.(\omega)) \in Z).$$

In other words, for $F \in \mathcal{F}$, $C \in \tilde{\mathcal{C}}$,

$$\tilde{P}^n(F \times C) = \int_F \delta_C(y(\omega)) dP^n(\omega).$$

We will show that the sequence $\tilde{P}^n$ is tight.

For a positive constant M , let $V_M = \{v \in \mathcal{A}^k[0, T] : \|v_T\| \leq M\}$. From Corollary 2.9 we know that V_M is a compact Borel subset of $\mathcal{A}^k[0, T]$. Proposition 3.4 implies that for any given $\varepsilon > 0$, there exists an M such that

$$(3.17) \quad P(\mathcal{D}^d[0, T] \times \mathcal{U} \times V_M) \geq 1 - \varepsilon$$

for every $P \in \mathcal{R}_{s,x}^\lambda$, $s \in [0, T]$, $x \in A$. Therefore, for the corresponding $\tilde{P}$, we have

$$(3.18) \quad \tilde{P}(\mathcal{D}^d[0, T] \times \mathcal{U} \times V_M \times C^d[0, T]) \geq 1 - \varepsilon.$$

By Proposition 3.3 we know that the probability laws of y with respect to P^n , denoted by $\bar{P}^n$, are relatively compact, and there exists a compact subset $K \subset C^d[0, T]$ with

$$\bar{P}^n(K) \geq 1 - \varepsilon,$$

or equivalently,

$$(3.19) \quad P^n(\omega : y.(\omega) \in K) \geq 1 - \varepsilon \quad \forall n.$$

From Proposition 3.1, (3.19) may be written as

$$(3.20) \quad \tilde{P}^n(\Omega \times K) \geq 1 - \varepsilon \quad \forall n.$$

We now consider the coordinate process x . Let

$$D = K + G(V_M) = \{y + G(v), y \in K, v \in V_M\}.$$

By Lemma 2.10 we know that $G(V_M)$ is a relatively compact subset of $\mathcal{D}^d[0, T]$ under the pseudopath topology. Since the uniform topology is stronger than the pseudopath topology, K is also a compact subset in $\mathcal{D}^d[0, T]$, hence so is D . From Proposition 3.1 we have

$$(3.21) \quad \tilde{P}^n(D \times \mathcal{U} \times V_M \times K) = \tilde{P}^n(\mathcal{D}^d[0, T] \times \mathcal{U} \times V_M \times K).$$

Let $S = D \times \mathcal{U} \times V_M \times K$. Since $\mathcal{U}$ is a compact space, we know that S is a relatively compact subset in $\mathcal{Z}$. Moreover, from (3.18), (3.20), and (3.21) we have

$$\tilde{P}^n(S) \geq 1 - 2\varepsilon$$

for every n . Thus $\{\tilde{P}^n\}$ is a tight sequence of probability measures on $\mathcal{Z}$. By the Prohorov theorem there is a subsequence $\{\tilde{P}^{n_k}\}$ and a probability $\tilde{P}$ on $(\mathcal{Z}, \tilde{\mathcal{Z}})$ such that $\tilde{P}^{n_k} \rightarrow \tilde{P}$ weakly. Define

$$P = \tilde{P}|_\Omega,$$

i.e., in the terminology of Jacod and Mémmin [17], P is the Ω -marginal of $\tilde{P}$, then it is easy to see that $P^{n_k} \rightarrow P$ weakly. The proof of the theorem will be completed if $P \in \mathcal{R}_{s,x}^\lambda$. By Proposition 3.1 we need only to show that there exists a continuous process Y on $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$ such that

- (1) $P(Y. = x. - G.(v)) = 1$;
- (2) $P(x_r = x, v_r = 0, 0 \leq r \leq s) = 1$;
- (3) $\tilde{M}_t \phi \in \mathcal{M}_2^c \quad \forall \phi \in C_b^2(\mathbb{R}^d)$; and

$$(4) \quad J(s, P) \leq \lambda.$$

Note that (4) is obvious from Theorem 3.6. To show (1), note that the set $\{(\omega, y) : x(\omega) = y + G(v(\omega))\}$ is a closed subset of $\mathcal{Z} = \Omega \times C^d[0, T]$, and, therefore,

$$(3.22) \quad \tilde{P}(x = y + G(v)) \geq \limsup \tilde{P}^n(x = y + G(v)) = 1.$$

If we define $Y(\omega) = x(\omega) - G(v(\omega))$, then $\tilde{P}((\omega, y) : Y(\omega) = y) = 1$. Thus Y is a continuous process on $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, and (1) follows from (3.22). Moreover, $\{y(0) = x\}$ is closed in $\mathcal{Z}$, so

$$\tilde{P}\{y(0) = x\} \geq \limsup_k \tilde{P}^{n_k}\{y(0) = x\} = 1$$

or $P\{\omega : Y(0) = x\} = 1$. It follows that $x(0) = x$ almost surely (P).

For (2), let

$$B_m = \left\{ \omega = (x, \mu, v) : \|x(t) - x\| \leq \frac{1}{m}, v_t = 0, 0 < t \leq s - \frac{1}{m} \right\}.$$

It is easy to see that B_m is closed in Ω and $B_1 \supset B_2 \supset \cdots$, and

$$\bigcap_m B_m = \{\omega = (x, \mu, v) : x_t = x, v_t = 0, 0 < t \leq s\}.$$

Since $P^{n_k} \in \mathcal{R}_{s_{n_k}, x_{n_k}}$, we know for each m , $P^{n_k}(B_m) = 1$ for large k since $x_{n_k} \rightarrow x$. But $P^{n_k} \rightarrow P$ weakly, so

$$P(B_m) \geq \limsup_k P^{n_k}(B_m) = 1,$$

and, therefore,

$$\begin{aligned} & P\{\omega : x_t = x, v_t = 0, 0 \leq t \leq s\} \\ &= P\left(\bigcap_m B_m \cap \{x(0) = x\}\right) = \lim_m P(B_m) = 1. \end{aligned}$$

Finally, we prove (3). For any bounded continuous function $H(\cdot)$ on Ω , if we define

$$\tilde{H}(\tilde{\omega}) = H(\omega) \quad \forall \tilde{\omega} = (\omega, y) \in \Omega \times C^d[0, T],$$

then $\tilde{H}$ is a continuous function on $\mathcal{Z}$. For any fixed $t \geq s$, the function

$$z = (x, \mu, v, y) \in \mathcal{Z} \mapsto \bar{M}_t \phi(z) = \phi(y_t) - \int_s^t \tilde{\mathcal{L}} \phi(\theta, x_\theta, y_\theta, \mu_\theta) d\theta$$

is continuous on $\mathcal{Z}$. In fact, it can be shown as in the proof of Lemma 3.5 that the integral part of the function $\bar{M}_t \phi$ is continuous on $\mathcal{Z}$, and the continuity of the function

$$z = (x, \mu, v, y) \rightarrow \phi(y_t)$$

on $\mathcal{Z}$ follows from the fact that $C^d[0, T]$ is endowed with the uniform topology. Thus, for $0 \leq u < t \leq T$, the function

$$z = (x, \mu, v, y) \rightarrow \tilde{H}(z) [\bar{M}_t \phi(z) - \bar{M}_u \phi(z)]$$

is a bounded and continuous function on Z , and since $\tilde{P}^{n_k} \rightarrow \tilde{P}$ weakly, we have

$$(3.23) \quad \lim_k E^{\tilde{P}^{n_k}} \left\{ \tilde{H}[\tilde{M}_t \phi - \tilde{M}_u \phi] \right\} = E^{\tilde{P}} \left\{ \tilde{H}[\tilde{M}_t \phi - \tilde{M}_u \phi] \right\}.$$

Again, by (3.22) and the definition of Y , we have $\tilde{P}(\tilde{M}_t \phi = \tilde{M}_u \phi) = 1$, where $\tilde{M}_t \phi$ is defined by (3.2) with y replaced by Y . Thus (3.23) can be rewritten as

$$(3.24) \quad \lim_k E^{P^{n_k}} \left\{ H[\tilde{M}_t \phi - \tilde{M}_u \phi] \right\} = E^P \left\{ H[\tilde{M}_t \phi - \tilde{M}_u \phi] \right\}.$$

For every bounded continuous function H on Ω that is $\mathcal{F}_u$ -measurable, the left-hand side of (3.24) is zero since $\tilde{M} \phi \in \mathcal{M}_2^c$ on the filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$. By a routine limit procedure, we have

$$E^P[H(\tilde{M}_t \phi - \tilde{M}_u \phi)] = 0$$

for each bounded $\mathcal{F}_u$ -measurable function H . Thus $(\tilde{M}_t \phi, \mathcal{F}_t)$ is a martingale under P . The continuity of this martingale follows from that of Y . The proof is therefore complete. $\square$

We can now prove the main theorem of this paper.

THEOREM 3.8. *The control problem has an optimal solution, i.e., there exists a $P^* \in \mathcal{R}_{s,x}$ such that*

$$J(s, P^*) = \inf_{P \in \mathcal{R}_{s,x}} J(s, P).$$

Proof. By Proposition 2.11 and Theorem 3.7 we know that $\mathcal{R}_{s,x}^{\lambda(x)}$ is nonempty and compact. Moreover, it is obvious that

$$\inf_{P \in \mathcal{R}_{s,x}} J(s, P) = \inf_{P \in \mathcal{R}_{s,x}^{\lambda(x)}} J(s, P).$$

Now $J(s, \cdot)$ is a lower-semicontinuous function on $\mathcal{R}_{s,x}$, so it attains its minimum on the compact set $\mathcal{R}_{s,x}^{\lambda(x)} \subset M_1(\Omega)$, i.e., there exists a $P^* \in \mathcal{R}_{s,x}^{\lambda(x)} \subset \mathcal{R}_{s,x}$ such that

$$J(s, P^*) = \inf_{P \in \mathcal{R}_{s,x}^{\lambda(x)}} J(s, P) = \inf_{P \in \mathcal{R}_{s,x}} J(s, P),$$

which completes the proof. $\square$

Recall that the value function is defined by

$$W(s, x) = \inf_{P \in \mathcal{R}_{s,x}} J(s, P).$$

Let us define

$$\mathcal{R}_{s,x}^o = \{P \in \mathcal{R}_{s,x} : W(s, x) = J(s, P)\}.$$

By Theorem 3.8, $\mathcal{R}_{s,x}^o \neq \emptyset$ for any $(s, x) \in \Sigma$. It can be easily verified that it is a compact subset of $M_1(\Omega)$.

Before we prove the measurability of the value function W , we give a result which is used in Haussmann and Suo [14]. We adopt the notations of Stroock and Varadhan [31, Chap. 12].

LEMMA 3.9. *The map $\mathcal{R}^o : \Sigma \mapsto \text{comp}(\mathbb{M}_1(\Omega))$ is Borel measurable. Moreover, there exists a measurable selector H of $\mathcal{R}^o$, i.e., $H(s, x) \in \mathcal{R}_{s,x}^o \ \forall (s, x) \in \Sigma$ and $H : \Sigma \mapsto \mathbb{M}_1(\Omega)$ is Borel measurable.*

Proof. By [31, Lem. 12.1.8], we need only to show the following: for $(s_n, x_n) \in \Sigma$, $s_n \rightarrow s$, $x_n \rightarrow x$, $P^n \in \mathcal{R}_{s_n, x_n}^o$, there exists a subsequence P^{n_k} and $P \in \mathcal{R}_{s,x}^o$ such that $P^{n_k} \rightarrow P$.

Since $x_n \rightarrow x$, we may assume $\lambda(x_n) \leq \lambda$ for some constant λ . Therefore $\{P^n\} \subset \{\mathcal{R}_{s,x}^\lambda, (s, x) \in [0, T] \times A\}$ with $A = \{x, x_1, x_2, \dots\}$ a compact set. By Theorem 3.7, there exists a subsequence P^{n_k} and $P \in \mathcal{R}_{s,x}$ such that $P^{n_k} \rightarrow P$. From Theorem 3.6, it can be seen easily that $P \in \mathcal{R}_{s,x}^o$. The measurability of $\mathcal{R}^o$ is thus proved.

The existence of a measurable selector H is a consequence of [31, Thm. 12.1.10].

□

COROLLARY 3.10. *$W(\cdot, \cdot)$ is a Borel-measurable function.*

Proof. From Theorem 3.6 we know the map $(s, x, P) \mapsto J(s, P)$ is lower semi-continuous and thus Borel measurable. The corollary follows from the fact that $W(s, x) = J(s, H(s, x))$ is the composition of two Borel-measurable mappings. □

4. Some comments. (a) The model studied in this paper includes the case of the *monotone follower* problem as formulated in Karatzas [18] and Karatzas and Shreve [21] by letting $k = d$ and $g(\theta) = I$, $0 \leq \theta \leq T$, where I denotes the $d \times d$ unit matrix. Moreover, if we take $k = 2d$ and $g(\theta) = (I, -I)$, $0 \leq \theta \leq T$, then the model reduces the *bounded variation* problem as discussed in Chow, Menaldi, and Robin [6], among others.

We have assumed that $c^i(\cdot) > 0$. This condition is necessary for the existence of optimal controls to the general problem formulated in this paper (see the proof of Proposition 3.4). Thus it excludes the case of the so-called *cheap control* problems, i.e., $c(\cdot) = 0$. This type of problem is discussed in Chiarolla and Haussmann [4], [5] and Menaldi and Robin [25]. However, our method works for problems with *finite fuel* constraints, because for any $y \in \mathcal{D}^k[0, T]$, the set

$$\{v \in \mathcal{A}^k[0, T] : v^i(t) \leq y^i(t), \forall t, 1 \leq i \leq k\}$$

is closed in $\mathcal{A}^k[0, T]$. This follows from the fact that $v_n \rightarrow v$ in $\mathcal{A}^k[0, T]$, then $v_n^i(t) \rightarrow v^i(t)$ ($1 \leq i \leq k$) at all the continuity points of v^i . For the problems with finite fuel constraints, see Karatzas [20], Karatzas and El Karoui [8], and a more recent work by Bridge and Shreve [1] among others.

(b) Now we explain why the method used in this paper is not suitable to the problem where terminal costs are allowed. If we define $\Gamma_s(\omega) = \phi(x_T(\omega))$ for $\omega = (x, \mu, v)$ with ϕ a continuous function defined in $\mathbb{R}^d$, we cannot get the lower semicontinuity for Γ_s . The reason is that $x^n \rightarrow x$ in the pseudopath topology only ensures that $x^n(\cdot) \rightarrow x(\cdot)$ in Lebesgue measure, and, therefore, $x^n(T) \rightarrow x(T)$ may not hold.

But we can modify the formulation of the problem to allow a terminal cost, i.e., let

$$J(\alpha) \equiv E^P \left\{ \int_s^T f(t, x_t, u_t) dt + \int_{[s, T]} c(t) \cdot dv_t + \phi(x_{T+}) \right\}.$$

Note the additional term $c(T) \cdot \Delta v_T$ in the second integral and the relation $x_{T+} = x_T + g(T) \Delta v_T$. We can replace $\mathcal{D}^d[0, T]$ by $\mathcal{D}^d[0, T] \times \mathbb{R}^d$ and $\mathcal{A}^d[0, T]$ by $\mathcal{A}^d[0, T] \times \mathbb{R}^d$ in the canonical space to obtain the results if ϕ is lower semicontinuous.

(c) For the same reason as explained in (b) we cannot allow pointwise constraints of the type $\Phi(x(t_0)) = 0$ almost surely for some lower-semicontinuous function Φ on $\mathbb{R}^d$ (which may take the values $\pm\infty$) and fixed $t_0 \in [0, T]$. But it is seen easily that the following kind of integral constraint may be added to the problem:

$$\int_s^T f_0(t, x_t, u_t) dt + \int_s^T c_0(t) \cdot dv_t \leq 0 \quad \text{a.s.},$$

where f_0, c_0 satisfy the same conditions as f and c , except that the positivity of $c_0(\cdot)$ is relaxed to *bounded below*. Moreover, from the proof of Lemma 3.5 we can conclude that the following kind of constraint may also be added to the model:

$$\int_s^T f_1(t, x_t, u_t) dt + \int_s^T c_1(t) \cdot dv_t = 0$$

with $f_1(t, \cdot, \cdot), c_1(\cdot)$ continuous on Σ (for each $0 \leq t \leq T$), $[0, T]$, respectively. Of course, we must now assume the existence of an admissible control. See Haussmann and Lepeltier [13] for constraints of these types in the classical control problems.

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